

The cognitive turn in computational modeling for experimental pragmatics

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Abstract

concrete.

1 Experimental pragmatics and computational modeling

The linguistic subfield of pragmatics studies how language is used for efficient communication and how communicated meaning is modulated, if not created, in a particular context by systematic, context-general patterns of use. Its origins lie in the philosophy of language, in particular the analytical tradition of philosophy, in work on topics like *speech-acts* (Austin, 1962; Searle, 1969), *conversational implicature* (Grice, 1975), or *presupposition* (Stalnaker, 1973; Karttunen, 1973). Much early work on these topics remained invested in a logical-formal approach (Gazdar, 1979; Kadmon, 2001), but the early 2000's brought a deep and empowering incision: a widely adopted and ultimately very successful *experimental turn* (Noveck and Sperber, 2004; Noveck, 2018), in parallel to similar empirical re-orientations in other hitherto very analytically minded subfields of linguistics like formal syntax (Sprouse, 2007, 2023) and formal semantics (Meibauer and Steinbach, 2011; Cummins and Katsos, 2019).

The experimental turn in pragmatics was accompanied by formal and computational work from the start, with two main strands of approaches emerging at opposite ends of a spectrum. On one end of the spectrum, *grammaticalism* focuses on language as an abstract, computationally encapsulated system [MF: refs]. On the other end, *probabilistic pragmatics* seeks explanations of pragmatic phenomena in terms linguistic communication as a special form of social interaction, also taking aspects of domain-general cognition of language users into account [MF: refs]. Figure 1 gives a coarse-grained overview of the development of, in particular, probabilistic pragmatics as the culmination point of research that combined aspects of individual cognition, such as bounded rationality [MF: refs], and functional explanations based on optimal information transfer in communication [MF: refs]. While experimental data can mitigate between alternative explanations proposed in the grammaticalist tradition [MF: refs], the more socio-cognitive approach adopted by probabilistic pragmatics is much better suited to incorporate *both* linguistic theory *and* aspects of general cognition (such as perception, adaptation and decision-making) into a unified computational model of the *data-generating process*, i.e., the (cognitive) processes by which individual data observations in an experiment come about.

[MF: continue with something like: seen in this light computational modeling can serve a bridging function; one direction done so far, the other still needs work]

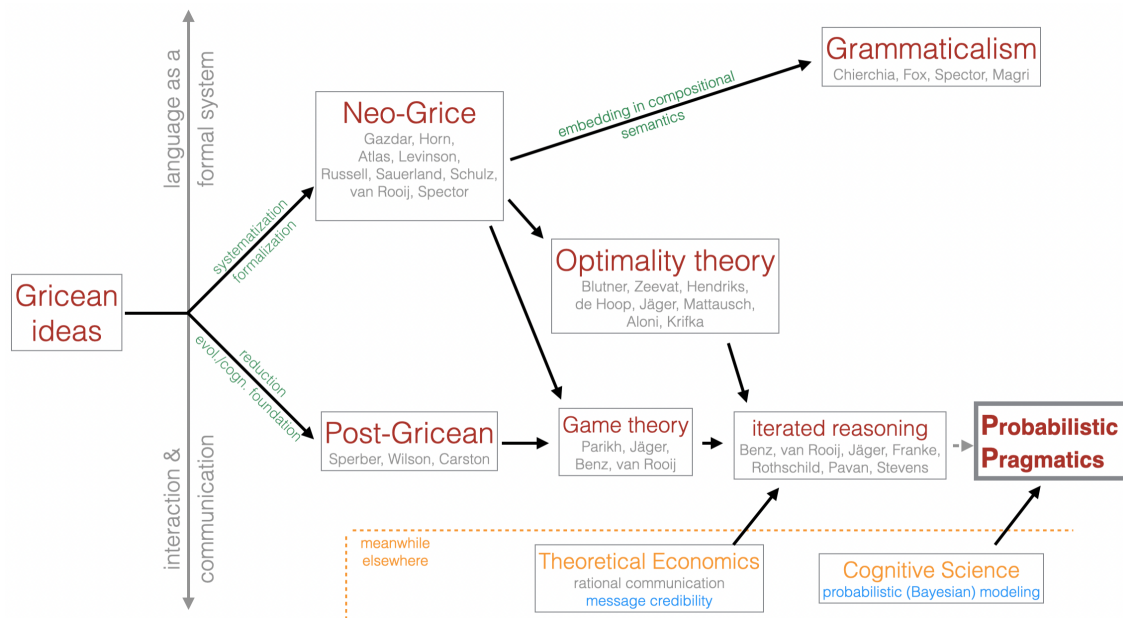


Figure 1: Genealogy of probabilistic pragmatics. [MF: describe what's pictured here (if you keep it; maybe make this figure and the text part of an InfoBox?)]

Section [MF: XYZ] elaborates on this idea of cognitive models acting as theory-driven statistical models for the observation-generating process. It covers some main lessons learned from probabilistic pragmatic modeling of experimental data, and points to key open questions for the future.

- two specific contributions of explicit (computational / formal) modeling in XPrag:
 1. **bridge to linguistic theory:** (a successful past)
 - give linguistic theory (e.g., latent concepts) an explicit role in our picture of the data-generating process via theory-driven bespoke modeling
 - experimental turn: experiments based on theory
 - bring in the theory as a first-order citizen → theory-driven statistical modeling
 2. **bridge to general cognition:** (a possible (bright) future)
 - frame pragmatic reasoning in the broader picture of human cognition
 - stuff for the future: proclaiming the *cognitive turn* in experimental pragmatics

2 Theory-driven probabilistic modeling

- modeling in general:
 - all models are wrong; some are useful
 - even if all models are wrong, we should prefer models that are less wrong in the sense of Isaak
 - so how our (less) wrong models useful?
 - while for data: garbage in — garbage out
 - for models: more assumptions in — more conclusions out
 - e.g., causal modeling
 - reasonable assumptions in; interesting conclusions out
 - all models contain some assumptions about the **data-generating process**
 - more assumptions about the nature of that process → more information gained from the data

3 Towards the cognitive turn

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